

3.

1. Discrete Kinematic model.

A glider moving b/w grid nodes
separated by a distance ΔL along
a unit direction vector n , the time
velocity over ground V_{ground} is the
vector sum of its through-water velocity
 V_g and the ocean current u .

$$V_{ground} \approx V_g(t_s) + u \cdot n.$$

$$\Delta t = \frac{\Delta L}{\min[V_g(t_s) + u \cdot n, V_{min}]}$$

2. Continuous path integral:

total travel time T from a starting point
 x_0 to destination x along path s is the
integral of the differential time $dt = ds / V_{ground}$

$$T_x = \int_s \frac{ds}{V_g(x) + u(x) \cdot n(s)}$$

$n(s) = \frac{ds}{ds}$ unit tangent vector to the
path.

3. Hamilton-Jacobi - Bellman principle of optimality.

$$T(x) = \min_n \left[T(x - \Delta L n) + \frac{\Delta L}{v_g + v \cdot n} \right]$$

First order Taylor expansion on $T(x - \Delta L n)$:

$$T(x - \Delta L n) \approx T(x) - \Delta L (\nabla T \cdot n)$$

Substitute in Bellman eqn.

$$T(x) = \min_n \left[T(x) - \Delta L (\nabla T \cdot n) + \frac{\Delta L}{v_g + v \cdot n} \right]$$

Subtract $T(x)$ from both sides & divide by ΔL .

$$0 = \min_n \left[-\nabla T \cdot n + \frac{1}{v_g + v \cdot n} \right].$$

4. Anisotropic eikonal equation:

$$\max_n [(\nabla T \cdot n)(v_g + v \cdot n)] = 1$$

because Speed depends on travel n ,

forms Anisotropic eikonal eqn.

$$v_g |\nabla T| + v \cdot \nabla T = 1$$

The "Reachability Front"

$$T(x) = T_{\text{horizon}}$$

59:

$$x_{k+1} = x_k + [u_k + v_{own,k}] \Delta t + w_k$$

where

- x_k is the true system state at time step k .
- u_k is the velocity input vector
- $v_{own,k}$ is the ownship velocity component
- w_k is the zero-mean white process noise vector, representing unmodeled disturbances with covariance $Q_k = E[w_k w_k^T]$.

Expected value (mean) of true state. we take mathematical expectation $E[\cdot]$ of state x_k

$$\hat{x}_{k+1} = E[x_{k+1}] = E[x_k + [u_k + v_{own,k}] \Delta t + w_k]$$

By linear property of expectation:

$$\hat{x}_{k+1} = E[x_k] + E[u_k + v_{own,k}] \Delta t + E[w_k]$$

Because w_k is zero-mean noise, $E[w_k] = 0$.

Substituting $E[x_k] = \hat{x}_k$ we get deterministic mean propagation eqn:

$$\hat{x}_{k+1} = \hat{x}_k + [u_k + v_{own,k}] \Delta t$$

Covariance (P_{k+1})
 The error covariance matrix P_k represents the expected squared error of our estimate.

Step A: Define error Dynamics.

Subtract mean propagation eqn from the true state eqn. to find the estimation error

$$e_{k+1} = x_{k+1} - \hat{x}_{k+1}$$

$$e_{k+1} = (x_k [u_k + v_{own,k}] \Delta t + w_k) - (\hat{x}_k + [\hat{u} + v_{own,k}] \Delta t)$$

$$e_{k+1} = (x_k - \hat{x}_k) + (u_k - \hat{u}) \Delta t + w_k$$

Let $e_k = x_k - \hat{x}_k$. In a linearized system (EKF), the propagation of the state error through the system dynamics is represented via the State Transition matrix (F):

$$e_{k+1} = F e_k + w_k$$

Step B: Compute the expected covariance.

the covariance matrix at step $k+1$ is

$$P_{k+1} = E[e_{k+1} e_{k+1}^T] \quad \text{Substitute the error eqn.}$$

$$P_{k+1} = E[(F e_k + w_k)(F e_k + w_k)^T]$$

$$P_{k+1} = E[F e_k e_k^T F^T + F e_k w_k^T + w_k e_k^T F^T + w_k w_k^T]$$

Step C: Apply expectation properties.

Distribute the expectation operator across each term.

$$1. E[F e_k e_k^T F^T] = F E[e_k e_k^T] F^T = F P_k F^T$$

$$2. E[w_k w_k^T] = Q_k \quad (\text{By definition of the process noise covariance})$$

$$3. E[F e_k w_k^T] = 0 \text{ and } E[w_k e_k^T F^T] = 0$$

This is because the process noise at step k (w_k) is completely uncorrelated with the current estimation error (e_k), which depends only on past noise.

Combining the remaining nonzero terms yields the final covariance time update:

$$P_{k+1} = F P_k F^T + Q_k$$

S5:

1. Optimization formulation.

- let $x_v(t)$ be the vessel trajectory and $u(t)$ be the control input.

Objective is to minimise the total route energy (or) cost E_{route} , subject to safety and scheduling chance constraints.

$$\min_{u(t)} \int_{t_0}^T \dot{E}(x_v(t), u(t), \text{environment}) dt$$

Subject to :

(i) Collision Avoidance: $P(\|x_v(t) - x_{ic}(t)\| R_{phy}) < 0.01$

(ii) Rendezvous/Arrival: $P_{\text{meet}}(x_v(t) \in x_{\text{target}}) \geq 0.9$

(iii) Dynamics: $\dot{x}_v = f(x_v, u)$.

where $R_{phy} = R_{\text{vessel}} + R_{\text{iceberg}}$ is the physical collision radius.

2. Derivation of Stand-off Distance.

the chance $P(\text{collision}) < 0.01$ is transformed into a deterministic geometric boundary (the stand off distance, $D_{\text{stand off}}$). The planned path must satisfy

$$\|x_v(t) - x_{ic}(t)\|_{\text{planned}} \geq D_{\text{stand off}}$$

Buffer must account for max closure
blw the vessel and the iceberg that
can occur at the 99th percentile of
uncertainty.

→ we decompose relative position
error into three spatial components

(a) Navigation uncertainty (σ_{nav})

(b) Kinematic propagation Error

(Iceberg response) ::

$$\Delta_{ice} = \int_{t_0}^{t_0 + t_{response}}$$

$$V_{iceberg}(t) dt \approx V_{iceberg} t_{response}$$

(c) Static Geometry & Residuals (M_{safety})

$$M_{safety} = r_{vessel} + r_{iceberg} + E_{radar}$$

where E_{radar} accounts for iceberg

size estimation error & sensor resolution

limits to us (ambiguity) of sensors at

various distances - stationarity is also

etc. (to handle the ambiguity of the sensor)

projected - lower of the two

$$||\hat{x}(t) - x(t)|| \leq \sigma_{nav}(t)$$

Synthesis

P_{phy} is less than 1%, the planned min distance must be strictly greater than the sum of the 99th percentile.

extreme values of the system's uncertainty

required deterministic constraint:

$$D_{\text{standoff}} = \sigma_{\text{nav}} + V_{\text{icubeg}} \cdot t_{\text{response}} + M_{\text{offset}}.$$

By enforcing

$$\|x_v(t) - x_{ic}(t)\|_{\text{planned}} \geq D_{\text{standoff}} \text{ in the}$$

Path planner, the original chance constraint

$P(\text{collision}) < 0.01$ is structurally

Satisfied.

Sg:

1. Heave Standard Deviation:

$$\sigma_h = k \cdot H_s$$

→ while amplitude of waves follow a Rayleigh distribution, the instantaneous surface elevation follows a zero mean Gaussian distribution.

→ H_s - significant wave height.

H_s is approximately equal to 4 times the standard deviation of the surface elevation ($H_s \approx 4\sigma$).

~~σ is proportional to H_s~~ so $\sigma_h = k H_s$
 k is around 0.25.

2. Single Attempt lock probability.

$$P_{lock,1} = \text{erf}\left(\frac{e}{\sqrt{2} \cdot \sigma_h}\right)$$

Because instantaneous heave position z is zero mean Gaussian random variable with variance σ_h^2 .

its PDF is

$$f(z) = \frac{1}{\sigma_h \sqrt{2\pi}} \exp\left(-\frac{z^2}{2\sigma_h^2}\right)$$

the probability of a successful lock on one attempt is the probability that z falls b/w $-e$ and e .

$$P_{lock,1} = \int_{-e}^e \frac{1}{\sigma_h \sqrt{2\pi}} \exp\left(-\frac{z^2}{2\sigma_h^2}\right) dz$$

$$P_{lock,1} = \frac{2}{\sigma_h \sqrt{2\pi}} \int_0^e \exp\left(-\frac{z^2}{2\sigma_h^2}\right) dz$$

We use standard deviation of error function: $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$

$$\text{let } t = \frac{z}{\sqrt{2}\sigma_h}$$

therefore, $dz = \sqrt{2}\sigma_h dt$

when $z = e$ the upper limit becomes $t = \frac{e}{\sqrt{2}\sigma_h}$

$$P_{lock,1} = \frac{2}{\sigma_h \sqrt{2\pi}} \int_0^{\frac{e}{\sqrt{2}\sigma_h}} \exp(-t^2) (\sqrt{2}\sigma_h) dt$$

the σ_h and $\sqrt{2}$ terms cancel out.

$$P_{lock,1} = \frac{2}{\sqrt{\pi}} \int_0^{\frac{e}{\sqrt{2}\sigma_h}} \exp(-t^2) dt$$

$$P_{lock,1} = \text{erf}\left(\frac{e}{\sqrt{2} \cdot \sigma_h}\right)$$

3. Multiple Attempts Probability:

$$P_{\text{lock},n} = 1 - (1 - P_{\text{lock},1})^n$$